

CTMT Compatibility Lock and Holonomy Obstruction: A Čech–Bundle Theorem, Logical Lock, and Synthetic Falsification Battery

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Abstract

This paper isolates the globalization problem for CTMT from the physical realization problem. The local axiom is that admissible CTMT morphisms are automorphisms of the local resolution geometry

$$\mathcal{G} = (\mathcal{O}; R, N, \Sigma), \quad R = \operatorname{im} J, \quad N = \ker J^T,$$

or of the conformal variant $\mathcal{G}_c = (\mathcal{O}; R, N, [\Sigma])$ when the protocol discards one global covariance scale. The compatibility problem is then classical in form: local charts glue when their transition morphisms are admissible and satisfy the Čech cocycle condition. This is the same logical pattern underlying vector-bundle and principal-bundle gluing theorems. The CTMT-specific content is the structure group: instead of an arbitrary $GL(n)$, transitions take values in $\operatorname{Aut}(\mathcal{G})$, i.e. the automorphism group of resolution geometry.

The paper distinguishes three levels that must not be conflated. First, edge compatibility means $\phi_{ij}^* \mathcal{G}_j = \mathcal{G}_i$. Second, weak globalization as a CTMT bundle requires the cocycle condition on triple overlaps. Third, strong globalization as a single trivial chart, or identity-monodromy closure, requires the resulting cocycle/holonomy class to be trivial. Nontrivial holonomy is therefore not automatically a failure of globalization; it is an obstruction to trivialization and a classifier of nontrivial global CTMT geometries. A final synthetic falsification battery confirms the logical lock: pairwise admissibility alone fails; cocycle closure succeeds; gauge-equivalent cocycles remain equivalent; nontrivial flat holonomy survives as a global class; Fisher-only compatibility gives false positives; and the strict/conformal covariance-scale valve behaves as predicted.

Contents

1	Strategic separation	2
2	Local CTMT geometry and admissible morphisms	2
3	Compatibility data	2
4	The compatibility gluing theorem	3
5	Holonomy and obstruction classes	3
6	Logical lock	4
7	Final synthetic falsification battery	4
8	Non-claims	5

1 Strategic separation

The CTMT compatibility problem has two different meanings.

Mathematical existence. Given local CTMT charts and transition morphisms, when does a global CTMT object exist?

Physical realization. Do real systems provide charts and transitions satisfying those hypotheses?

Only the first question is addressed here. It is a pure gluing problem. Classical geometry has long treated analogous problems through atlases, vector bundles, principal bundles, transition functions, Čech cocycles, and holonomy [1–4].

Non-claim 1 (No proof that Nature satisfies the hypotheses). No theorem in this paper proves that coils, fluids, turbine machinery, seismic catalogues, quantum systems, or spacetime satisfy CTMT compatibility hypotheses. Experiments can test those hypotheses only after the mathematical target is stated.

2 Local CTMT geometry and admissible morphisms

Definition 2 (Resolution geometry). A local CTMT chart over an open set U_i carries a resolution geometry

$$\mathcal{G}_i = (\mathcal{O}_i; R_i, N_i, \Sigma_i),$$

where $\mathcal{O}_i$ is the local observation space, $R_i = \text{im } J_i$ is the resolved sector, $N_i = \ker J_i^T$ is the unresolved sector, and Σ_i is the covariance structure declared by the observation protocol.

Axiom 3 (Local morphism axiom). A local map $\phi : \mathcal{O}_i \rightarrow \mathcal{O}_j$ is CTMT-admissible iff it preserves resolution geometry:

$$\phi \in \text{Mor}_{\text{CTMT}}(i, j) \iff \phi^* \mathcal{G}_j = \mathcal{G}_i.$$

In representation form this requires

$$\phi(R_i) = R_j, \quad \phi(N_i) = N_j, \quad \phi \Sigma_i \phi^T = \Sigma_j,$$

or the conformal replacement $\phi \Sigma_i \phi^T = \lambda \Sigma_j$, $\lambda > 0$, when the protocol works with $[\Sigma]$.

3 Compatibility data

Let $\{U_i\}_{i \in I}$ be an open cover. Write $U_{ij} = U_i \cap U_j$ and $U_{ijk} = U_i \cap U_j \cap U_k$.

Definition 4 (CTMT transition datum). A CTMT transition datum consists of admissible overlap maps

$$\phi_{ij} : \mathcal{G}_j|_{U_{ij}} \longrightarrow \mathcal{G}_i|_{U_{ij}}$$

satisfying the inverse condition

$$\phi_{ii} = I, \quad \phi_{ji} = \phi_{ij}^{-1}.$$

Definition 5 (Cocycle closure). A transition datum is a CTMT cocycle if, on every nonempty triple overlap,

$$\boxed{\phi_{ij} \phi_{jk} \phi_{ki} = I.} \tag{1}$$

This is the direct CTMT analogue of the usual cocycle condition for vector bundles and principal bundles [1–3].

Remark 1 (Why this is the lock point). Edge compatibility only says that every pair of neighboring charts agrees. The cocycle condition says that all pairwise agreements are mutually consistent on triple overlaps. This is exactly the difference between local pairwise admissibility and an honest glued object.

4 The compatibility gluing theorem

Theorem 1 (Weak CTMT globalization). Let $\{(U_i, \mathcal{G}_i)\}$ be local CTMT charts. Suppose there are transition maps ϕ_{ij} such that:

- (a) $\phi_{ij}^* \mathcal{G}_i = \mathcal{G}_j$ on U_{ij} ;
- (b) $\phi_{ji} = \phi_{ij}^{-1}$;
- (c) $\phi_{ij} \phi_{jk} \phi_{ki} = I$ on U_{ijk} .

Then the local charts glue to a global CTMT resolution-geometry bundle. The glued object is unique up to CTMT isomorphism induced by 0-cochain gauge transformations.

Proof 6 (Proof sketch). Form the disjoint union of the local trivial pieces and impose the equivalence relation generated by identifying $(x, v)_j \sim (x, \phi_{ij}(x)v)_i$ on overlaps. Reflexivity and symmetry follow from $\phi_{ii} = I$ and $\phi_{ji} = \phi_{ij}^{-1}$. Transitivity on triple overlaps is exactly the cocycle condition. The quotient inherits local trivializations and the transition functions take values in $\text{Aut}(\mathcal{G})$. This is the standard vector-/principal-bundle gluing construction with structure group reduced to the CTMT automorphism group [1–3].

Corollary 7 (Pairwise compatibility is insufficient). Admissibility of every overlap map does not imply weak globalization. The triple-overlap cocycle condition is an independent requirement.

5 Holonomy and obstruction classes

Definition 8 (CTMT holonomy of a chart loop). For a chart loop $\gamma = (i_0, i_1, \dots, i_p = i_0)$, define

$$H_\gamma = \phi_{i_0 i_1} \phi_{i_1 i_2} \cdots \phi_{i_{p-1} i_p}.$$

If every edge is CTMT-admissible, then

$$H_\gamma \in \text{Aut}(\mathcal{G}_{i_0}).$$

Proposition 9 (Holonomy lives in the CTMT automorphism group). For any admissible closed chart loop γ , the loop product H_γ is an automorphism of the base chart’s resolution geometry.

Proof 10. Each transition is an isomorphism of resolution geometries. A composition of such isomorphisms around a closed loop starts and ends at the same chart, hence is an automorphism of that chart. This is the same categorical mechanism by which classical parallel transport around loops yields holonomy elements in the structure group [3, 4].

Theorem 2 (Strong trivialization obstruction). Assume a weak global CTMT bundle has flat locally constant transition data. It admits a single global CTMT trivialization over the chosen cover iff the transition cocycle is a coboundary, equivalently iff its holonomy representation is trivial up to the declared gauge equivalence.

Proof 11 (Proof sketch). A global trivialization is a family of local gauges $a_i : U_i \rightarrow \text{Aut}(\mathcal{G}_i)$ such that

$$\phi_{ij} = a_i^{-1} a_j.$$

This is precisely the statement that the Čech 1-cocycle is a coboundary. For flat locally constant transition data, the same information is encoded by the monodromy/holonomy representation of loops into the structure group. Trivial monodromy gives a global gauge; nontrivial monodromy obstructs a single identity-monodromy chart, even though the global bundle may exist [1, 2, 4].

Remark 2 (Important correction). Nontrivial holonomy is not automatically failure of globalization. It is failure of trivialization or identity-monodromy closure. The weaker global CTMT bundle may still exist and may be classified by its holonomy/cocycle class.

6 Logical lock

The lock is the following forced chain:

local morphism axiom $\Rightarrow \text{Aut}(\mathcal{G}) \Rightarrow \text{transition group} \Rightarrow \text{Čech cocycle} \Rightarrow \text{global CTMT bundle} / \text{obstruction class}.$
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It is a lock because weakening the data breaks known tests, while strengthening it recovers a classical gluing theorem.

Corollary 12 (Minimal compatibility stack). A CTMT compatibility proof requires exactly the following layers:

- (i) edge admissibility: $\phi_{ij}^* \mathcal{G}_i = \mathcal{G}_j$;
- (ii) inverse consistency: $\phi_{ji} = \phi_{ij}^{-1}$;
- (iii) cocycle closure: $\phi_{ij} \phi_{jk} \phi_{ki} = I$;
- (iv) gauge equivalence of transition data;
- (v) holonomy/cocycle class as the obstruction to strong trivialization.

Removing any layer either loses the glued object or confuses nontrivial global geometry with failure.

7 Final synthetic falsification battery

The following battery attacks the lock by separating edge admissibility, cocycle closure, gauge equivalence, nontrivial flat holonomy, Fisher-only compatibility, and covariance-scale alternatives.

ID	Scenario	Measured object	Value	Expected	Verdict
T0a	exact coboundary atlas	max edge residual	1.20e-15	~ 0	PASS
T0b	exact coboundary atlas	max triangle cocycle residual	1.46e-15	~ 0	PASS
T1a	pairwise ok, triple bad	defected edge residual	2.65e-16	~ 0	PASS EDGE
T1b	pairwise ok, triple bad	triangle holonomy $\ H - I\ _F$	5.657	> 0	FAIL COCYCLE
T1c	pairwise ok, triple bad	triangle holonomy $\ H + I\ _F$	1.25e-15	~ 0	NONTRIVIAL
T2	gauge equivalent cocycle	max cocycle residual after gauge	1.35e-15	~ 0	PASS

ID	Scenario	Measured object	Value	Expected	Verdict
T3a	nontrivial flat holonomy	monodromy edge residual	0	~ 0	PASS EDGE
T3b	nontrivial flat holonomy	$\ M - I\ _F$	2.0	> 0	NONTRIVIAL CLASS
T3c	nontrivial flat holonomy	eigenvalue count -1	1	> 0	CLASSIFIER
T4	bad edge mixing	d_N	0.277	$> \text{tol.}$	FAIL EDGE
T4b	bad edge mixing	strict covariance residual	0.071	$> \text{tol.}$	FAIL EDGE
T5a	Fisher false positive	relative Fisher change	0	0	FISHER PASS
T5b	Fisher false positive	strict covariance residual	1.414	> 0	CTMT FAIL
T5c	Fisher false positive	conformal covariance residual	0.943	> 0	CTMT FAIL
T6a	strict/conformal separator	strict residual, $4\Sigma \rightarrow \Sigma$	3.0	> 0	STRICT FAIL
T6b	strict/conformal separator	conformal residual, $4\Sigma \rightarrow \Sigma$	0	~ 0	CONFORMAL PASS
T6c	strict/conformal separator	recovered conformal scale	4.0	4	PASS
T7a	sector scale candidate	strict residual sector scale	3.536	> 0	STRICT FAIL
T7b	sector scale candidate	conformal residual sector scale	1.061	> 0	CONFORMAL FAIL
T7c	sector scale candidate	interpretation	–	new axiom	OPEN

Empirical finding 1 (Battery conclusion). The battery supports the exact logical separation required by the theorem. Pairwise admissibility can pass while the cocycle fails; gauge transformations preserve valid cocycles; nontrivial holonomy is a global class rather than an edge failure; Fisher-only criteria admit false positives; global covariance scale is the strict/conformal separator; and sector-relative scaling requires an explicit new axiom.

8 Non-claims

Non-claim 13 (No universal data claim). The synthetic battery is not evidence that a real system satisfies CTMT. It is an attack on the internal logic of the compatibility theorem.

Non-claim 14 (No claim that all holonomy is bad). Nontrivial holonomy is not automatically a defect. It is an obstruction to strong trivialization and may classify nontrivial global CTMT geometries.

Non-claim 15 (No replacement of established bundle theory). The gluing theorem is not new mathematics. The CTMT-specific contribution is the resolution-geometry structure group and the operational path from sensor samples to local chart automorphisms.

Non-claim 16 (No Fisher-only compatibility). Fisher invariance is a derived necessary condition under strict admissibility, not a sufficient condition for chart compatibility.

9 Conclusion

The compatibility problem now has a defensible mathematical form. Local CTMT geometry supplies a structure group $\text{Aut}(\mathcal{G})$. Compatible overlap maps are transition functions with values in this group. Weak globalization is standard Čech gluing. Strong trivialization is obstructed by the cocycle/holonomy class. The lock is therefore not empirical and not metric-only:

$\text{CTMT compatibility} = \text{admissible transitions valued in } \text{Aut}(\mathcal{G}) + \text{Čech cocycle closure.}$

The remaining physical question is whether real systems generate charts satisfying these hypotheses. That question is experimental. The mathematical lock is the gluing/obstruction theorem described above.

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